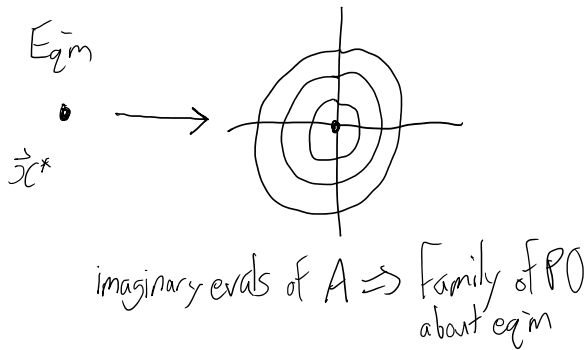


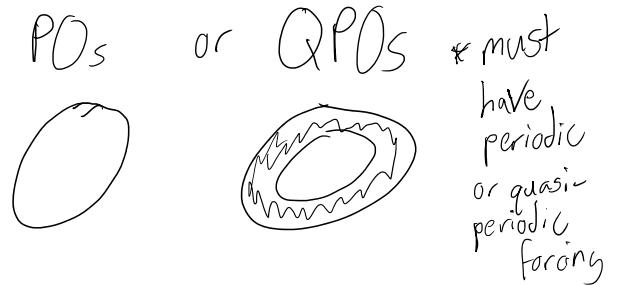
Goal: Develop a procedure to identify & compute regular sol'n's to understand allowable motion

Last Time:

"Simplest" Sol'n: Autonomous ($H(\vec{x}; \vec{p})$)



Non-autonomous ($H(\vec{x}, t; \vec{p})$)



Now, we want to compute PQs far away from eqm in auto. sys. & general PQs in non-autonomous systems

Single-Shooting

Find $\vec{z} = \begin{bmatrix} \vec{x}_0 \\ T \end{bmatrix}$ such that $f(\vec{z}) = \Phi_T(\vec{x}_0) - \vec{x}_0 \equiv 0$

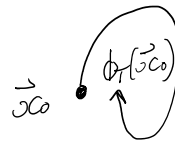
Our problem boils down to solving a root-finding problem: $f(\vec{z}) = \vec{0}$

→ Newton-Raphson update:

i) Start with an initial guess for $\vec{x}_0$ & T {

- auto. sys.
comes from linear analysis about the eqm
- periodic, non-auto. sys.
comes from eqm in the auto. sys. & period of the forcing OR resonant PQs in the auto. sys

ii) Predict: Integrate $\vec{x}_0$ for T time units



iii) Correct: Update $\vec{x}_0$ & T based on the local dynamics

Let $\vec{z}^{IC} = \begin{bmatrix} \vec{x}_0^{IC} \\ T^{IC} \end{bmatrix}$ where $f(\vec{z}^{IC}) \neq \vec{0}$. Assume $\vec{z}^{True}$, where $f(\vec{z}^{True}) = \vec{0}$, is pretty close $\vec{z}^{IC}$ such that $\vec{z}^{True} = \vec{z}^{IC} + \Delta\vec{z}$. Then,

$$f(\vec{z}^{True}) = \vec{0} \Rightarrow f(\vec{z}^{IC} + \Delta\vec{z}) = \vec{0} \Rightarrow f(\vec{z}^{IC}) + \left. \frac{\partial f}{\partial \vec{z}} \right|_{\vec{z}^{IC}} \Delta\vec{z} + \text{H.O.T.s} = \vec{0}$$

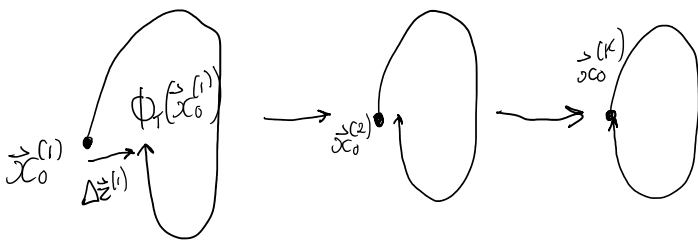
$$\Rightarrow f(\vec{z}^{IC}) + \left. \frac{\partial f}{\partial \vec{z}} \right|_{\vec{z}^{IC}} \Delta\vec{z} = \vec{0} \Rightarrow \Delta\vec{z} = - \left(\left. \frac{\partial f}{\partial \vec{z}} \right|_{\vec{z}^{IC}} \right)^T \left[\left. \frac{\partial f}{\partial \vec{z}} \right|_{\vec{z}^{IC}} \left. \frac{\partial f}{\partial \vec{z}} \right|_{\vec{z}^{IC}}^T \right]^{-1} f(\vec{z}^{IC})$$

→ Because we truncated H.O.T.s, $\Delta\vec{z}$ is an approximation & we must iterate to find our true root $\vec{z}^{True}$:

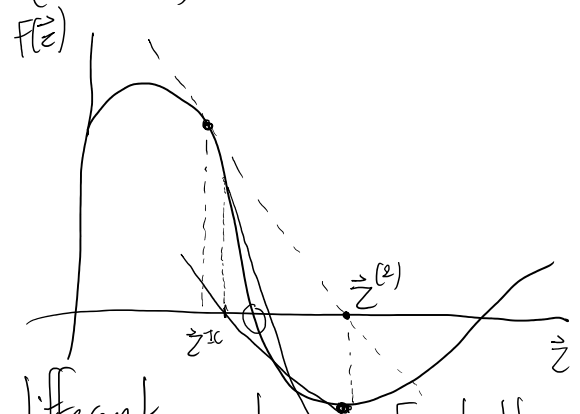
$$\vec{z}^{(i+1)} = \vec{z}^{(i)} - \left. \frac{\partial f}{\partial \vec{z}} \right|_{\vec{z}^{(i)}}^T \left[\left. \frac{\partial f}{\partial \vec{z}} \right|_{\vec{z}^{(i)}} \left. \frac{\partial f}{\partial \vec{z}} \right|_{\vec{z}^{(i)}}^T \right]^{-1} f(\vec{z}^{(i)})$$

→ In our case: $\frac{\partial f}{\partial \vec{z}} = \frac{\partial (\phi_T(\vec{x}_0) - \vec{x}_0)}{\partial (\vec{x}_0, T)} = \begin{bmatrix} \Phi(T, t_0) - \mathbf{I} & \dot{\mathbf{x}}(T) \end{bmatrix}$

iv) Repeat steps i) - iii) until $\|f(\vec{z}^{(i)})\| < \epsilon$ ($\sim 10^{-9}$)

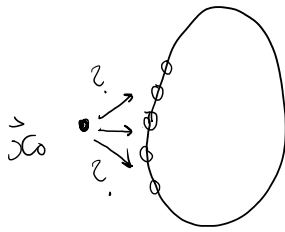


where



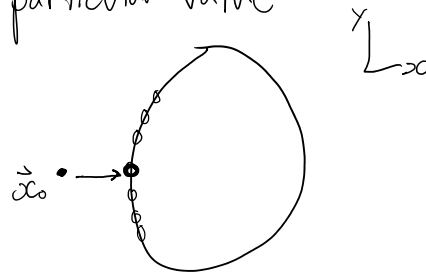
→ Single-shooting is an art! We can add different constraints for better convergence of the root-finding problem

e.g. Add a phase constraint: Note that any state on a PO satisfies our root-finding problem. This may "confuse" our Newton-Raphson update, as there is

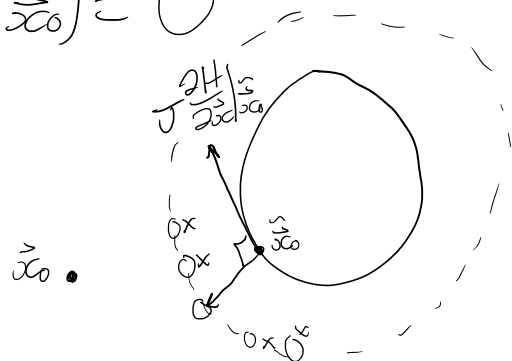


root-finding problem. This may confuse our Newton-Raphson update, as there is no single sol'n to $f(\vec{z}) = 0$

We can introduce a phase constraint $g(\vec{z})$ to lock down a single sol'n
 → Fix a component of $\vec{x}_0$ to a particular value
 $g(\vec{z}) = y_0 = 0$



→ Fix $\vec{x}_0$ based on the orientation of a previously computed PO, given by $\tilde{\vec{x}}_0$

$$g(\vec{z}) = \mathcal{J} \frac{\partial H}{\partial \vec{c}} \bigg|_{\tilde{\vec{x}}_0}^T (\vec{x}_0 - \tilde{\vec{x}}_0) = 0$$


We then augment our root-finding problem:

$$h(\vec{z}) = \begin{bmatrix} f(\vec{z}) \\ g(\vec{z}) \end{bmatrix} = \vec{0} \quad \text{where } \vec{z} = \begin{bmatrix} \vec{x}_0 \\ \tau \end{bmatrix}$$

$$\Rightarrow \vec{z}^{(i+1)} = \vec{z}^{(i)} - \left(\frac{\partial h}{\partial \vec{z}} \bigg|_{\vec{z}^{(i)}} \right)^{-1} f(\vec{z}^{(i)})$$

Summary so far: Single-shooting allows us to correct arbitrary POs, provided we have a good initial guess & the orbit isn't too large.
 In auto. sys., if our initial guess is from linearization about an eq'm pt, won't we correct onto a PO close to the eq'm?

How do we find POs that are "far away"?

We want to "continue" a PO computed from shooting techniques in a Hamiltonian system.

→ In auto. sys., POs exist in 1-parameter families. This parameter p can be something like the "radius" of PO or its orbit period T or even its energy $H(\vec{c}_0; \vec{p})$

→ Once we have computed a PO with a parameter p^* , we may want to compute a neighbouring PO with a parameter $p^* + \Delta p$

→ Again, this boils down to a root-finding problem

Natural Parameter Continuation

→ We introduce an additional parameter constraint into our shooting framework from earlier

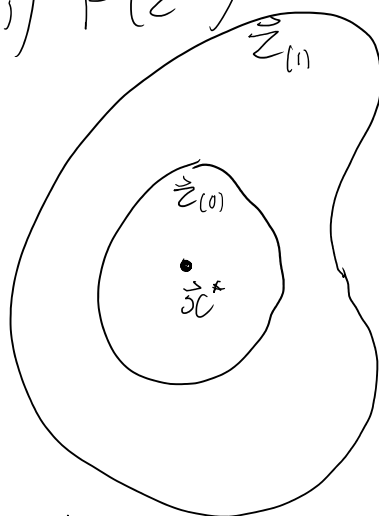
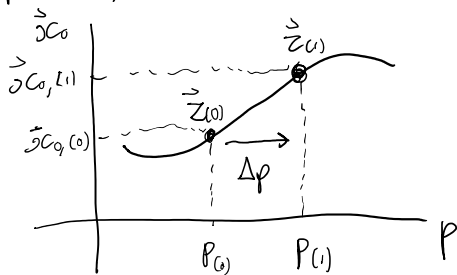
$$K(\vec{z}) = p - (p^* + \Delta p) = 0$$

The root-finding problem becomes,

$$F(\vec{z}) = \begin{bmatrix} f(\vec{z}) \\ K(\vec{z}) \end{bmatrix} = \vec{0} \quad \text{OR} \quad F(\vec{z}) = \begin{bmatrix} f(\vec{z}) \\ g(\vec{z}) \\ K(\vec{z}) \end{bmatrix} = \vec{0}$$

$$\Rightarrow \vec{z}^{(i+1)} = \vec{z}^{(i)} - \left(\frac{\partial F}{\partial \vec{z}} \bigg|_{\vec{z}^{(i)}} \right)^{-1} F(\vec{z}^{(i)})$$

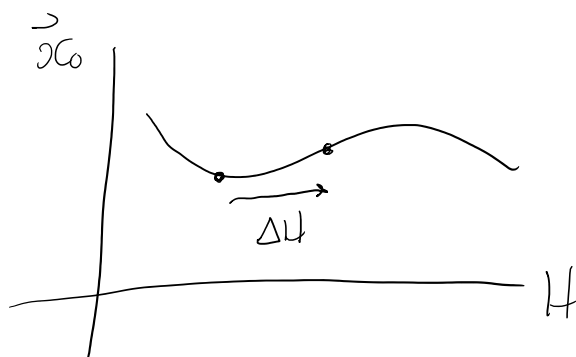
→ Graphically,



→ Example: Energy $H(\vec{x}_0; \mu)$ in autonomous sys.

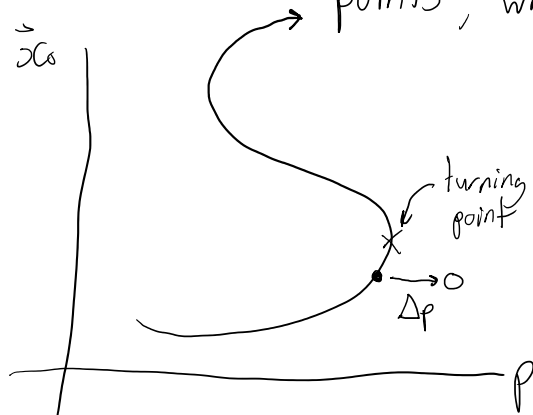
$$K(\vec{z}) = H - (H^* + \Delta H) \rightarrow \frac{\partial K}{\partial \vec{z}} = \frac{\partial}{\partial (\vec{x}_0, T)} (H - (H^* + \Delta H))$$

$$= \begin{bmatrix} \frac{\partial H}{\partial \vec{x}_0} \big|_{\vec{x}_0} & 0 \end{bmatrix} \quad J\bar{J} = -I$$



→ Natural parameter continuation provides POs at larger energies

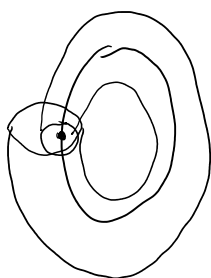
→ Disadvantages: Natural param. continuation breaks down at "turning points", where the 1-param. family "turns" around



→ To tackle turning points, we can use pseudoarclength continuation!

④ Analyze neighbourhoods of space about POs (stability of PO)

→ Recall that the STM Φ maps local deviations in phase space



→ Define the monodromy matrix as the STM integrated for one orbit period along a PO:

$$M = \Phi(T, 0)$$

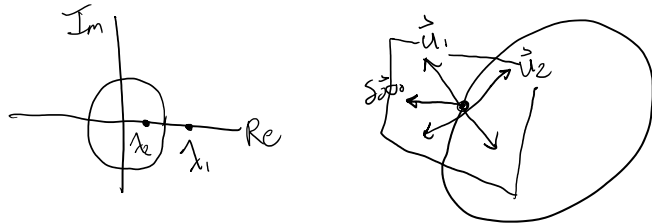
→ Note that M is a symplectic matrix, so its eigenvalues come in reciprocal pairs

Then case of evals for M (Floquet Theory)

Two Case of evals for M' (Floquet Theory)

Saddle

$$\lambda_1(M) = a \in \mathbb{R}, \lambda_2 = \frac{1}{a} \in \mathbb{R}$$



let $\vec{S}_{x0} = \vec{u}_1$, then

$$\vec{S}_{xT} = M \vec{S}_{x0} = \lambda_1 \vec{S}_{x0}$$

$$\vec{S}_{x2T} = M \vec{S}_{xT} = \lambda_1^2 \vec{S}_{x0}$$

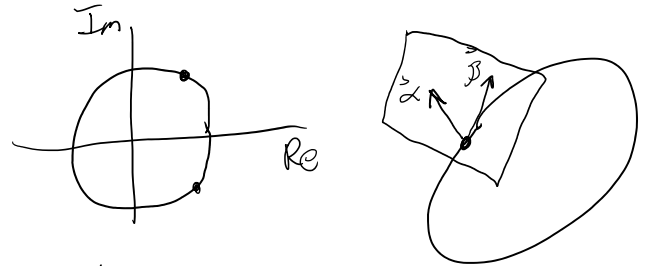
unstable manifold $\left\{ \vec{S}_{xKT} = \lambda_1^K \vec{S}_{x0} \xrightarrow{K \rightarrow \infty} \infty \text{ for } \lambda_1 > 1 \right.$

Similarly, $\vec{S}_{x0} = \vec{u}_2$, then

stable manifold $\left\{ \vec{S}_{xKT} = \lambda_2^K \vec{S}_{x0} \xrightarrow{K \rightarrow \infty} \vec{0} \text{ for } \lambda_2 < 1 \right.$

Center

$$\lambda_1 = a + ib, \lambda_2 = a - ib \text{ where } |\lambda_1| = |\lambda_2| = 1$$



let $\vec{u}_1 = \vec{\alpha} + i\vec{\beta}$

$$M^K \vec{u}_1 = \lambda_1^K \vec{u}_1 \text{ where } \lambda_1^K \text{ is also on the unit circle}$$

→ Magnitude of $\vec{u}_1$ is neither growing nor shrinking

→ Any $\vec{S}_{x0}$ along $\vec{\alpha}$ or $\vec{\beta}$ remains bounded about the PO (gives QPOs about a PO)

QPOs about Eq'm

→ "Step" into more than one center plane

$$\vec{S}_{x0} = \sum_{j=1}^{22} [\vec{S}_j \vec{v}_j] \begin{bmatrix} \cos \phi_j & \sin \phi_j \\ -\sin \phi_j & \cos \phi_j \end{bmatrix} \begin{bmatrix} C_j' \\ C_j'' \end{bmatrix}$$